Spatial Context for Wildfire Ignition Prediction

Project type: Research-flavor

1 Problem statement

This project asks whether conditions at multiple spatial scales improve seven-day wildfire ignition prediction compared with conditions within a single 5 km by 5 km cell. Wildfire ignition reflects interacting weather, drought, land cover, seasonal patterns, and historical fire activity. Accurate risk estimates could help prioritize monitoring and preparedness across the contiguous United States.

Environmental and historical predictors provide useful information, but a local cell may omit relevant surrounding vegetation, developed areas, and recent fire activity. The project will test whether this broader context adds predictive value. Each observation is a grid-cell and prediction-date pair. Its target equals one if at least one fire greater than 0.5 acres occurs in that cell during the following seven days, and zero otherwise. The outcome concerns qualifying ignition events, rather than fire spread, size, severity, or intensity.

2 Project goal and motivation

We will build a spatial and temporal prediction pipeline using PostgreSQL/PostGIS and XGBoost. The initial research idea is to keep the prediction cell fixed while progressively adding neighborhood features at 10, 20, and 50 km scales. The intended contribution is an experimental framework that identifies whether, and for which feature groups, broader context improves prediction beyond a local-only model. This contribution will be evaluated empirically; improved performance is a hypothesis, not an assumed result.

The project will compare local-only, local plus 10 km, local plus 10 and 20 km, and local plus 10, 20, and 50 km models. We will assess whether any gains justify additional processing and model complexity. Outputs will include continuous risk probabilities and odds. A current-risk heatmap of the lower 48 states is an eventual application, subject to current data availability and computing capacity.

3 Preliminary methodology

Data preparation will use the existing database to align records by grid_id and prediction_date. Predictors will include daily precipitation and temperature, recent dryness, weekly U.S. Drought Monitor severity and drought history, Annual NLCD land-cover percentages, seasonality, and historical fire counts and burned area over short and multiyear windows. Weather will come from the closest station with the required observations. Features must use only information available by the prediction date.

We will define neighborhood geometry consistently before extracting features and aggregate appropriate variables using counts, percentages, means, or extrema. The main spatial experiments will expand land-cover and fire-history context; drought context will be explored where supported. Weather and precipitation will remain tied to each cell’s assigned station, limiting conclusions about broader weather patterns. We will tune XGBoost depth, learning rate, tree count, sampling, and regularization under a consistent tuning budget across models.

3 Preliminary methodology continued

Evaluation will use nonoverlapping chronological periods, provisionally 2000–2016 for training, 2017–2020 for validation, and 2021–2023 for testing, subject to source coverage. Seven-day target windows that cross split boundaries will be excluded. Validation will guide model selection, and the test period will remain untouched until final evaluation. Long-history features will require earlier records or an explicit warm-up period.

Because ignition is rare, training will retain all qualifying positives and sample approximately three negatives per positive. Validation and testing may use approximately five negatives per positive if full-population evaluation is infeasible. We will record sampling rates and use representative evaluation data or sampling weights so reported performance reflects population prevalence. Sampled probabilities will receive a prevalence adjustment and calibration assessment before being interpreted as real-world risk; odds equal $p/(1 - p)$.

The primary metric will be PR-AUC, supplemented by ROC-AUC, precision, recall, F1, Brier score, and log loss. Calibration plots and risk-decile analysis will compare predicted risk with observed frequency. Feature-group ablations will isolate land-cover, fire-history, and feasible drought contributions. Comparisons will use identical evaluation observations, with uncertainty assessed through temporal block resampling where feasible. Feature-extraction time, training time, and memory use will help assess the cost of added spatial context.

Milestones and proposed schedule

Weeks 1–2: Audit the existing database, confirm source coverage and access, finalize neighborhood definitions, and validate target construction and temporal splits. Weeks 3–4: Extract local predictors, implement negative sampling, and train the 5 km baseline. Weeks 5–6: Build neighborhood features and tune the four spatial configurations. Week 7: Complete held-out evaluation, calibration checks, and ablation studies. Week 8: Consolidate results, document limitations and reproducible procedures, and prepare the final report and a heatmap prototype if feasible.

4 Required resources

The project requires the wildfire records, weather observations, U.S. Drought Monitor data, and Annual NLCD layers gathered by Tre Pepper, accessed through the project database and source downloads. We will verify historical coverage, update frequency, and usage terms before modeling. Software will include PostgreSQL/PostGIS, Python, XGBoost, and data-processing and evaluation libraries. A multicore computer with sufficient memory and storage will support initial experiments; larger national runs may require university computing resources, subject to access.

The full grid-by-date dataset and multiscale spatial joins may be expensive. Indexed queries, cached feature tables, negative sampling, and regional pilots will help control cost. Missing station records, incomplete fire histories, and delayed land-cover or fire reporting may reduce usable coverage and constrain deployment. Historical evaluation must respect source availability dates to avoid using information unavailable at prediction time.

5 Workload distribution

Tre Pepper has gathered the data sources and built the database used by the project. I will build the data-mining portion, develop the model, and run training and evaluation. We will coordinate through a shared feature specification and version-controlled scripts, with weekly reviews of data quality, database-to-model integration, and milestone progress. Tre will lead database and source clarification, while I will lead model comparisons and interpretation; we will jointly review the final report.